Lec-26

14/02/26 12:40 PM

8 9

digit

I I → Pattern

I
4 7 9
I

I D
1 5 3 D
I

I D
4 7 4

1234
I I I

I D

123

132

I I I I I
1 2 3 4 5 6

153 ✓
143 ✓
192 ✓
1

4 5 7 6
4 5 7 9

D D D D
5 4 3 2 1

7

1 to 7
2 to 8
3 to 7

D D I D I D
output

3	2	1	5	4	7	6

1 to 7

D D I D I D

[smaller digit]

3	2	1	5	4	7	6
0	1	2	3	4	5	6

I D D I D I

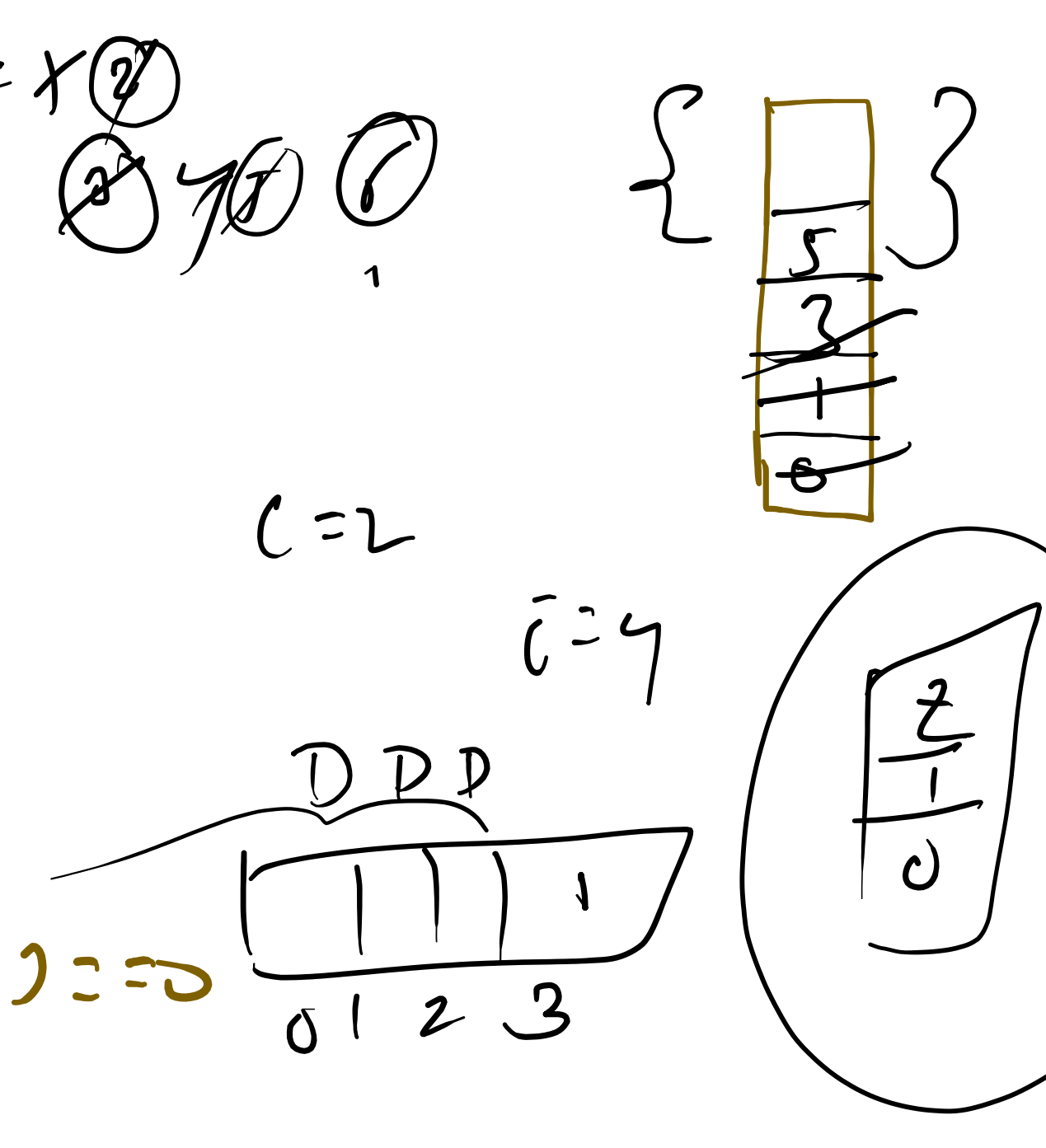
n+1

1	4	3	2	6	5	7
0	1	2	3	4	5	6

0 1 2 3 4 5
D D I D I D

i = r

c = 1 2


5
3
1
0
D D D
0 1 2 3
2
1
0

3	2	1	5	4	7	6
0	1	2	3	4	5	6

c=2

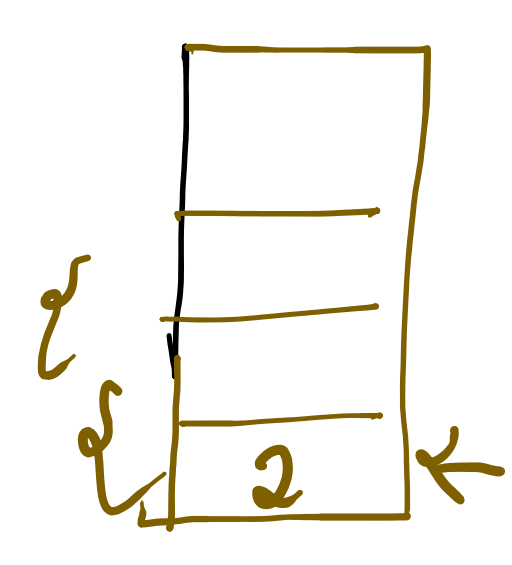
i=4

if(arr[cn][i]==1 || arr[i][cn]==0)

	0	1	2	3
0	0	1	1	1
1	1	0	1	1
2	0	0	0	0
3	1	1	1	0

O(n)

N=4



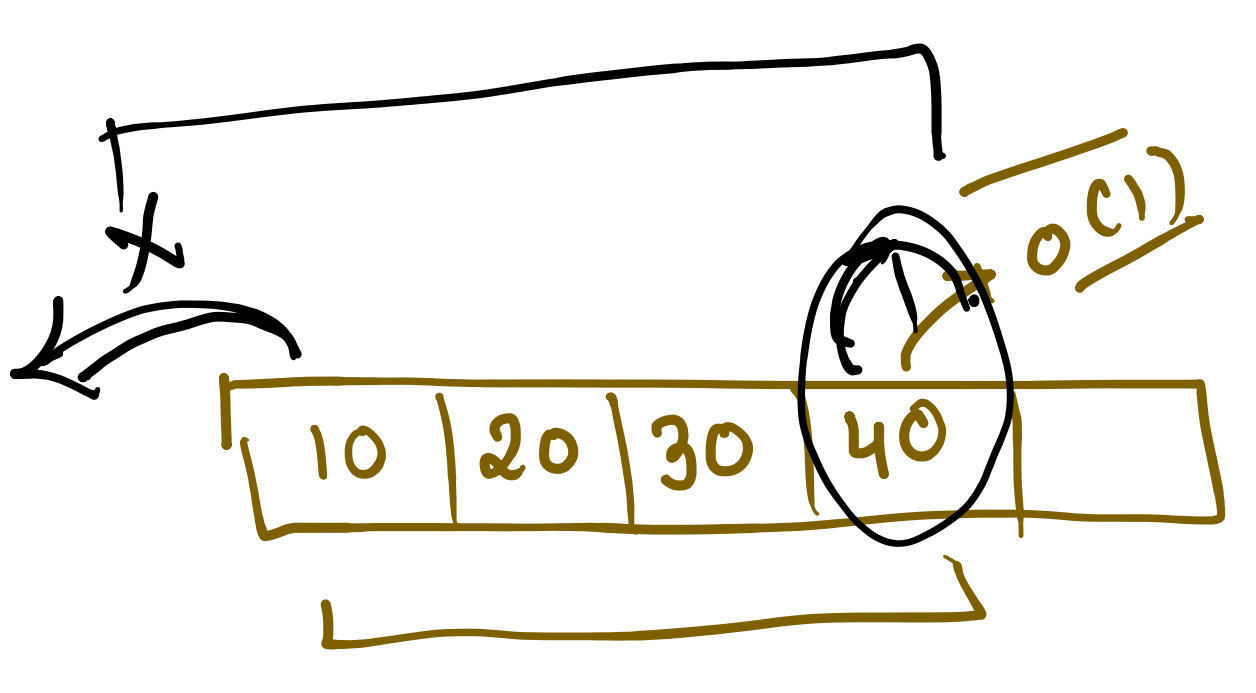
a=3
b=2
a=2
b=1
a=2
b=3

2 0, 2 1, 2 3 = 1

0 2
1 2
3 2 } 0

Stack → queue

push
pop
peek
O(1)

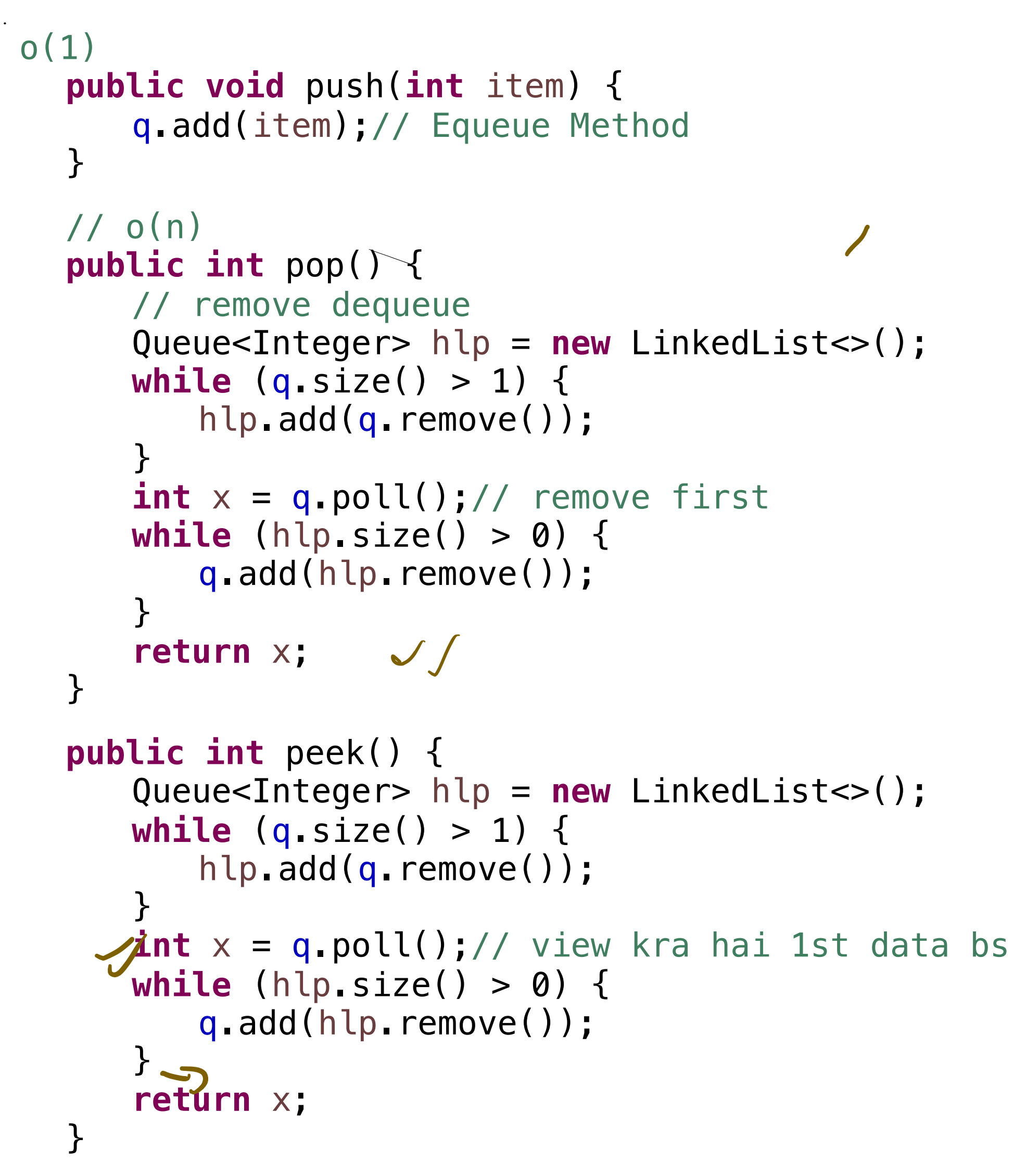

10
20
30
40
O(1)

Queue → stack

Deq

q.add(item) → push

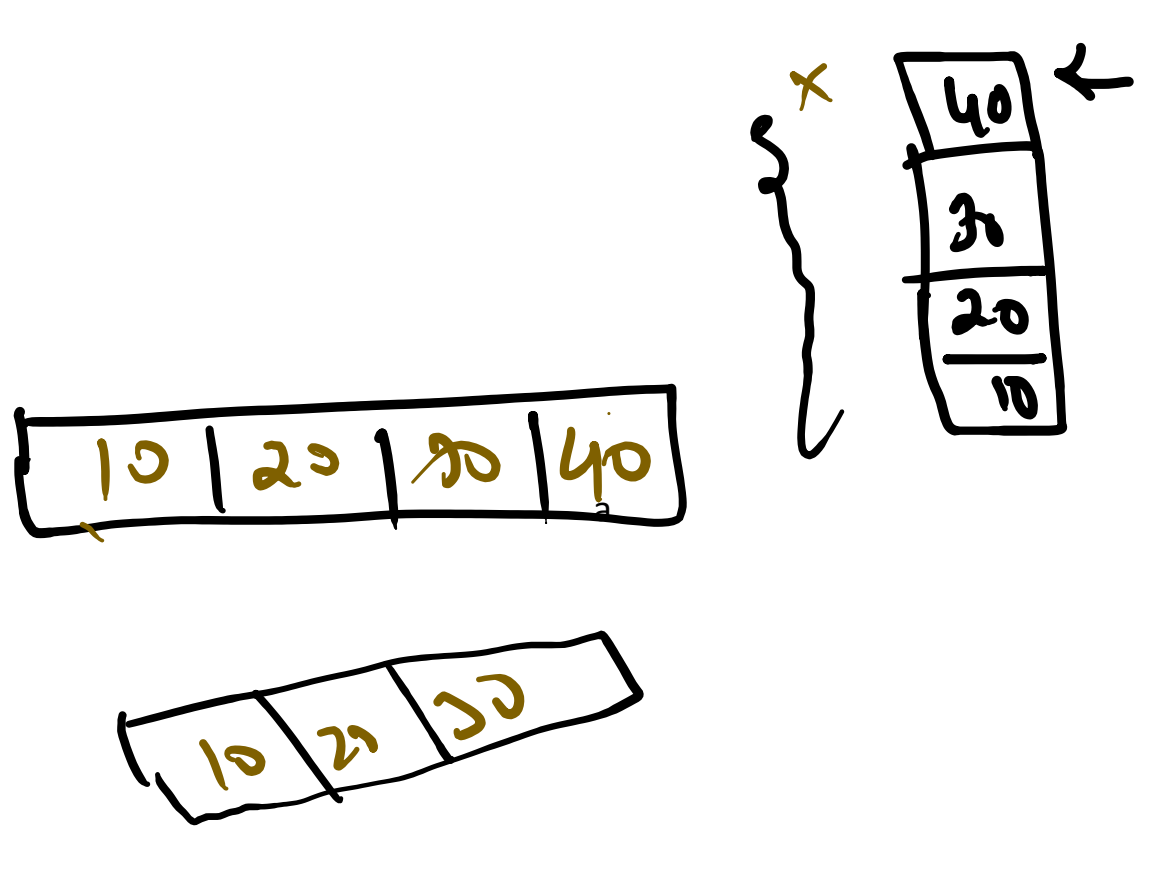
q.rev() →

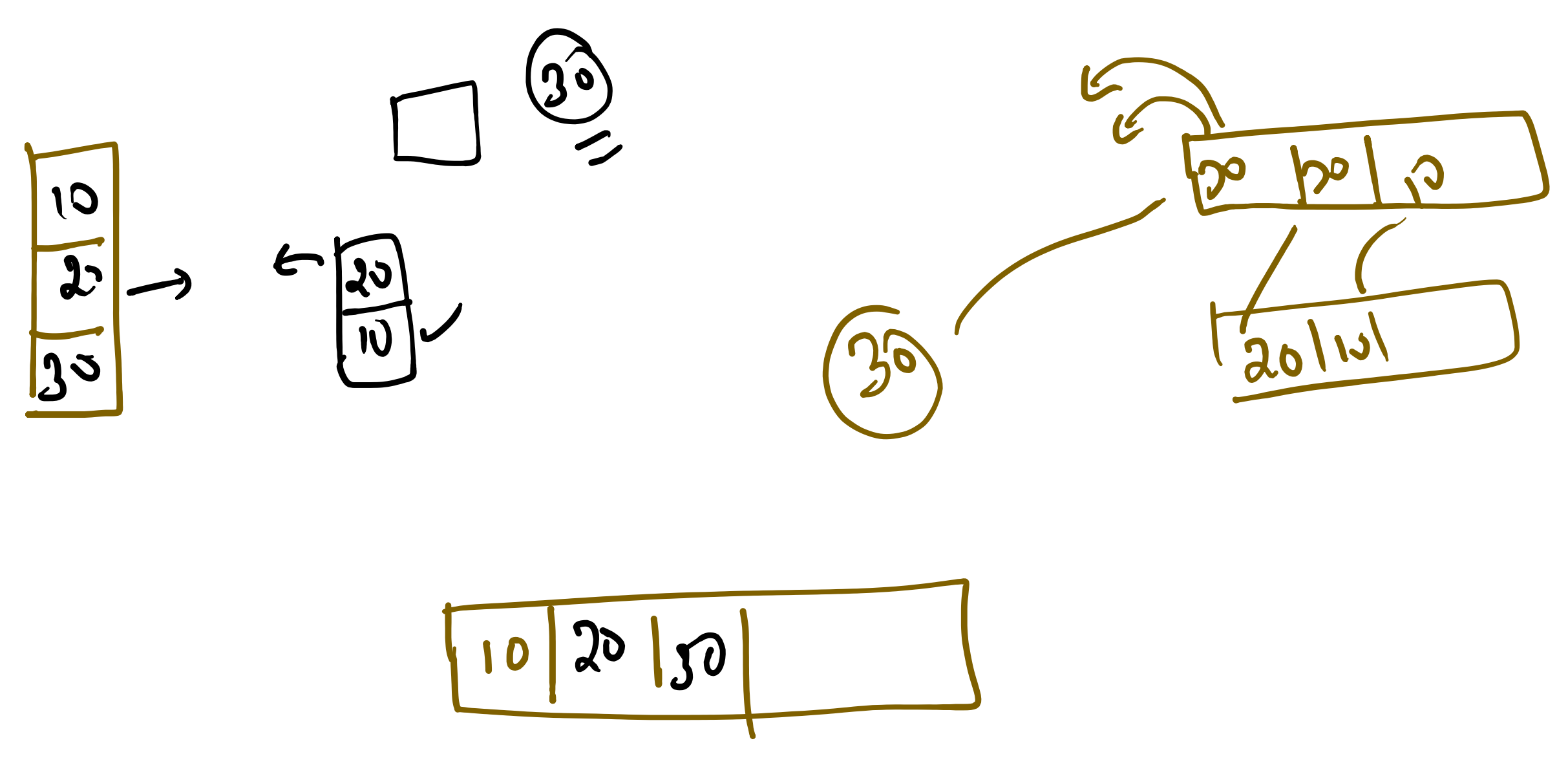


```
o(1)
public void push(int item) {
    q.add(item);// Equeue Method
}

// o(n)
public int pop() {
    // remove dequeue
    Queue<Integer> hlp = new LinkedList<>();
    while (q.size() > 1) {
        hlp.add(q.remove());
    }
    int x = q.poll();// remove first
    while (hlp.size() > 0) {
        q.add(hlp.remove());
    }
    return x;
}

public int peek() {
    Queue<Integer> hlp = new LinkedList<>();
    while (q.size() > 1) {
        hlp.add(q.remove());
    }
    int x = q.poll();// view kra hai 1st data bs
    while (hlp.size() > 0) {
        q.add(hlp.remove());
    }
    return x;
}
```

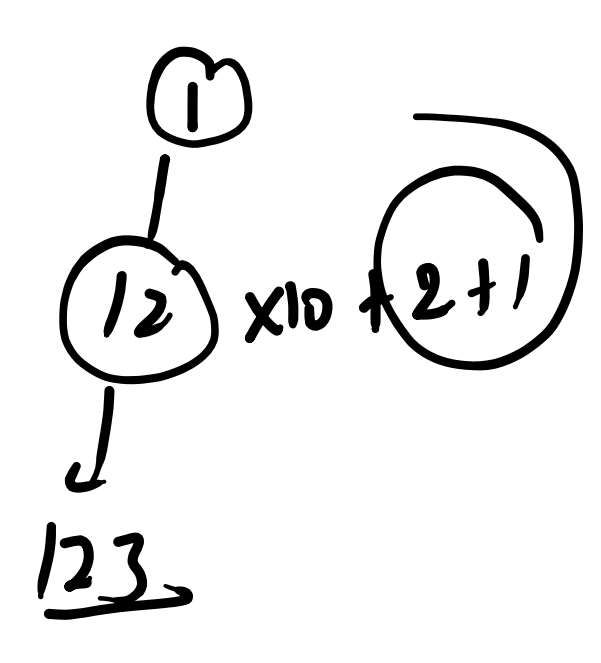

40
30
20
10
10
20
30
40
10
20
30


10
20
30
20
10
30
30
30
20
10
20
10
10
20
30

123

(1×10)(1+1)

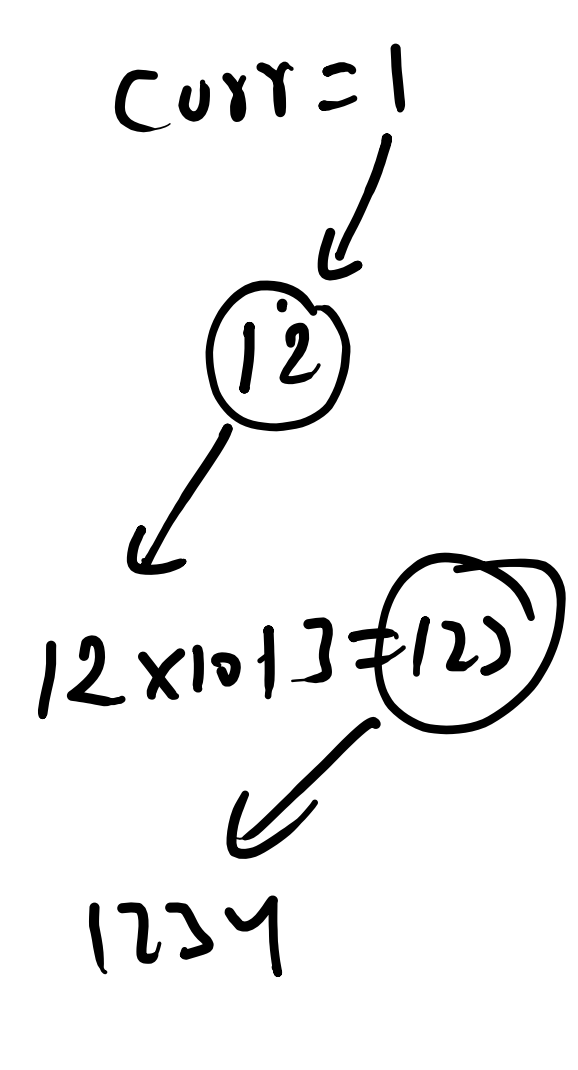
for(i=1 i<=9 i++)


1
12
×10 + 2+1
123

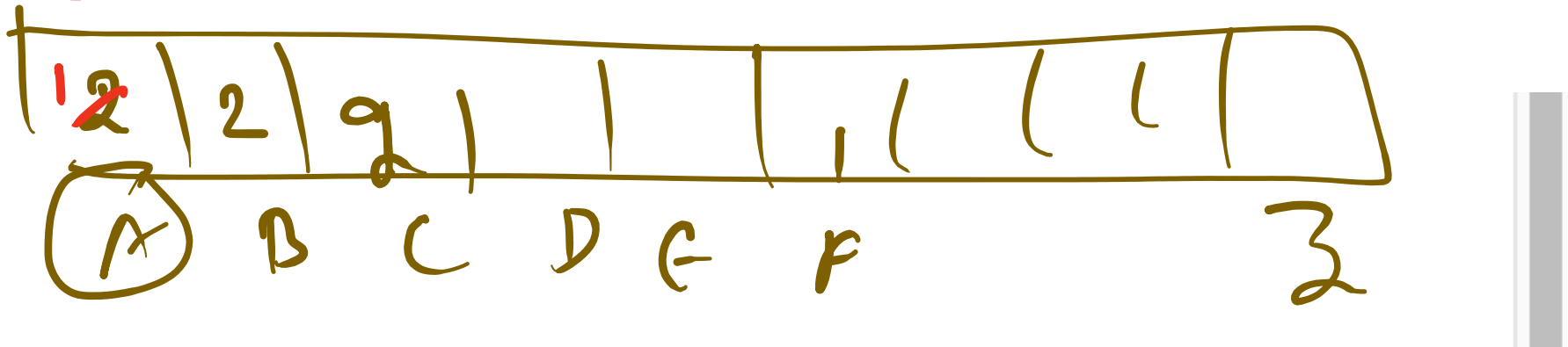
{low=100
high=300

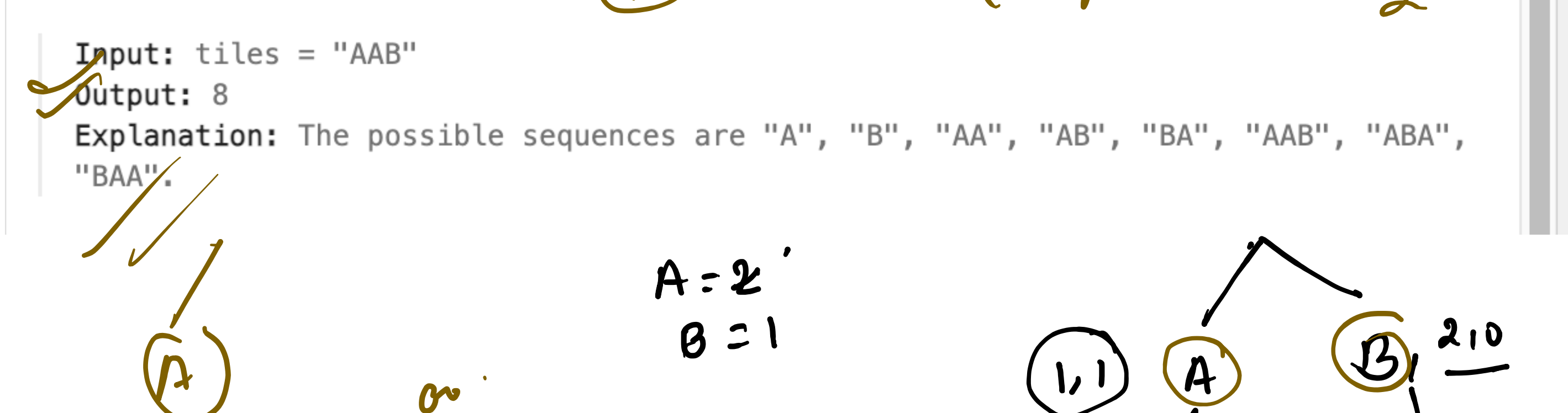
123×10+4

curr=1


12
12×10+3=123
1234

AABCBCF


1 2
2
9
A
B
C
D
E
F
3



Example 1:

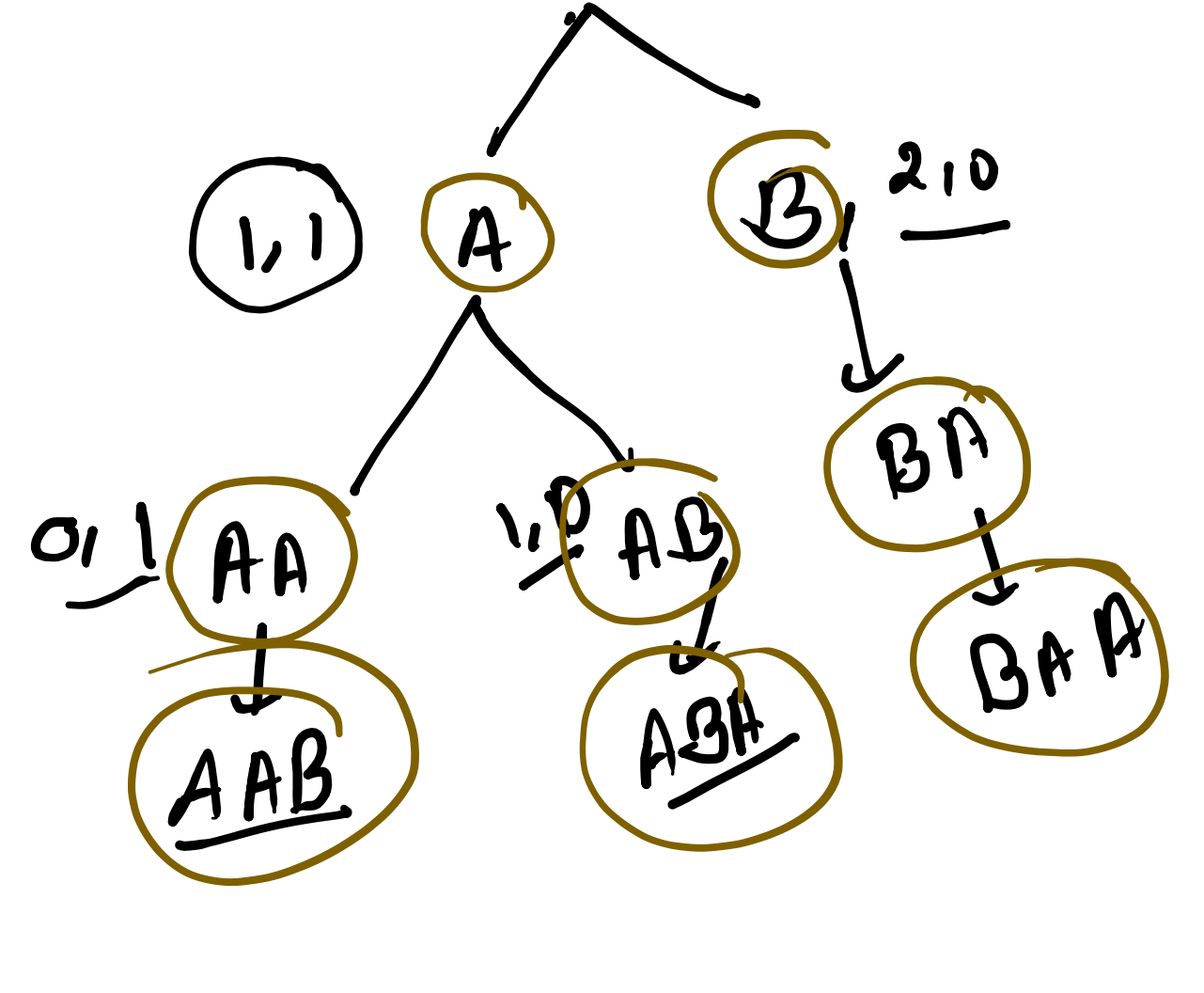
Input: tiles = "AAB"
Output: 8
Explanation: The possible sequences are "A", "B", "AA", "AB", "BA", "AAB", "ABA", "BAA".

A=2
B=1

A

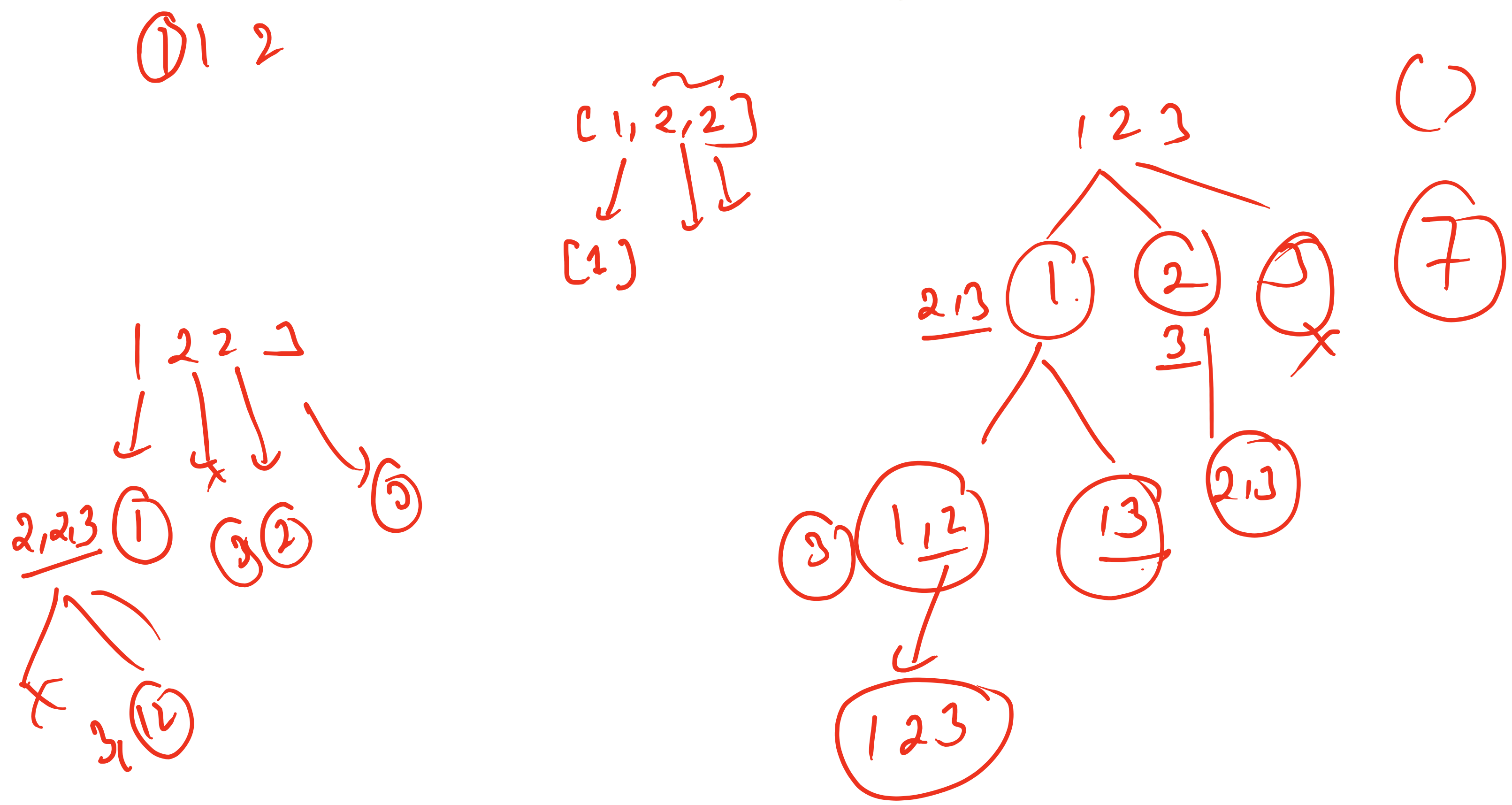
AA

0 1 2


1,1
A
B
2,1,0
0,1
AA
1,0
AB
BA
AAB
ABA
BAA

[1, 2, 2]

[1]


1 2 2 3
2,2,3
1
3
2
5
3
12
( )
7
1 2 3
2,3
1
2
3
3
2,3
3
1,2
13
123